

A Comparative Evaluation of Attitude Estimation Methods for a 6-DoF IMU

Anshul Rajee
Dept. of Robotics Engineering
Worcester Polytechnic Institute
Worcester, MA 01609
Email: arraje@wpi.edu

Imaan Haider
Dept. of Robotics Engineering
Worcester Polytechnic Institute
Worcester, MA 01609
Email: ihaider@wpi.edu

Zac Serocki
Dept. of Robotics Engineering
Worcester Polytechnic Institute
Worcester, MA 01609
Email: zserocki@wpi.edu

Abstract—This project presents a comparative evaluation of five attitude estimation methods applied to a 6-DoF Inertial Measurement Unit (IMU). A gyroscope integrator, an accelerometer estimator, a complementary filter, a Madgwick filter, and an unscented Kalman filter were implemented and assessed under identical conditions. The raw IMU measurements were first calibrated and then time-synchronized with Vicon measurements, which were used as ground truth. The roll, pitch, and yaw estimated by each method were compared against the Vicon orientation, and the root mean square error (RMSE) in each axis was used to quantify the relative accuracy of the five methods. The methods were then applied to four additional test datasets without Vicon ground truth to evaluate their behavior on unseen IMU data.

I. PROBLEM STATEMENT

Orientation cannot be reliably recovered from either inertial measurement sensor alone. Integrating gyroscope measurements yields smooth short-term estimates but accumulates drift over time, while the accelerometer provides a drift-free gravity reference that is corrupted by linear acceleration, and observes only roll and pitch. Attitude estimation methods differ in how they integrate the various measurements. The objective of this project was to implement five such methods: a gyroscope integrator, an accelerometer estimator, a complementary filter, a Madgwick filter, and an unscented Kalman filter, and to evaluate them under identical conditions. The IMU measurements were first calibrated by estimating the sensor biases and scale factors, and the resulting attitude estimates were compared against Vicon measurements as the ground truth. All five methods were evaluated on the same six provided train datasets and four test datasets, so that differences in performance can be attributed to the estimation method rather than the data.

II. IMU CALIBRATION AND DATA SYNCHRONIZATION

Before attitude estimation can be performed, the raw accelerometer and gyroscope measurements must be converted to physical units. Raw values are denoted with the subscript *raw* and calibrated values with a tilde.

The accelerometer biases B_a and scale factors S_a were provided beforehand, and are defined such that the scaled

measurement is expressed in units of g . The raw measurements were converted to m/s^2 using

$$\tilde{a}_x = (S_{a,x} a_{raw,x} + B_{a,x}) \times 9.81 \quad (1)$$

and the same calculation was repeated for a_y and a_z .

Since the IMU is stationary at the beginning of each dataset, the average of the first n raw gyroscope readings was used to estimate the gyroscope bias:

$$B_{\omega,x} = \frac{1}{n} \sum_{i=1}^n \omega_{raw,i,x} \quad (2)$$

where n is a tunable parameter depending on how long the IMU remains at rest before the trial. A value of $n = 200$ was chosen. The same method was applied to obtain $B_{\omega,y}$ and $B_{\omega,z}$. The scale factor S_ω of the gyroscope was provided beforehand as (3). Finally, we converted the raw gyroscope measurements to rad/s using (4).

$$S_\omega = \frac{3300}{1023} \times \frac{\pi}{180} \times 0.3 \quad (3)$$

$$\tilde{\omega} = S_\omega \times (\omega_{raw} - B_\omega) \quad (4)$$

The IMU and Vicon datasets were time-synchronized using a spherical linear interpolation (SLERP) algorithm written by Claude Opus 5 [1].

III. ATTITUDE ESTIMATION METHODS

Five attitude estimation methods were implemented using the calibrated and synchronized IMU measurements: a gyroscope integrator, accelerometer estimator, complementary filter, Madgwick filter, and unscented Kalman filter. The estimated orientations from each estimation method were then compared with the Vicon measurements.

Euler angles were initially used to represent and visualize the orientation in terms of roll, pitch, and yaw. However, Euler angles can reach singularities, which can result in gimbal lock. To avoid this issue, quaternions were used to represent the orientation and numerically integrate the measurements over time. The quaternion results were then converted back to Euler angles for plotting and comparison against Vicon readings.

A. Gyroscope Integrator

The gyroscope measures the angular velocity of the body about its x , y , and z axes. Over a single timestep, the angular velocity was treated as constant, so that the body rotates about the direction of the angular velocity vector by an angle equal to the angular rate multiplied by the timestep. This axis-angle pair was converted into a unit quaternion representing the incremental rotation over that timestep, which was then applied to the current orientation by quaternion multiplication. Repeating this at every timestep integrates the angular velocity into an orientation trajectory. Integration requires an initial condition, and any error in it propagates as a constant offset over the entire trajectory. The estimate was therefore initialized from the Vicon orientation at the first timestamp.

The magnitude and direction of the axis-angle pair are given by (5) and (6) respectively [2].

$$\alpha_\Delta = \|\vec{\omega}_t\| \Delta t \quad (5)$$

$$\vec{e}_\Delta = \frac{\vec{\omega}_t}{\|\vec{\omega}_t\|} \quad (6)$$

where Δt is the interval between consecutive IMU timestamps. Using these, we found the quaternion representing the incremental rotation and applied it to the current state quaternion using quaternion multiplication to get the next state quaternion [2].

$$q_\Delta = \left(\cos\left(\frac{\alpha_\Delta}{2}\right), \vec{e}_\Delta \sin\left(\frac{\alpha_\Delta}{2}\right) \right) \quad (7)$$

$$q_{t+1} = q_t \otimes q_\Delta \quad (8)$$

If the angular velocity magnitude was close to zero ($\leq 10^{-12}$), the rotation axis in (6) would be undefined. In this case the increment was treated as the identity rotation and the current state quaternion q_t was returned as q_{t+1} .

The resulting quaternions were converted to Euler angles to plot the roll, pitch, and yaw for comparison with the Vicon measurements.

B. Accelerometer Estimator

The accelerometer measures the linear acceleration (including gravity) acting on the body along its x , y , and z axes. When the IMU undergoes minimal acceleration, the accelerometer measures only the reaction to gravity. The direction of this measured gravity vector can therefore be used to estimate the tilt of the sensor. Rotation about the vertical axis is not observable with gravity alone, so only roll and pitch are observable from the accelerometer.

The measured specific-force vector is

$$\mathbf{a}_t = [a_x \ a_y \ a_z] \quad (9)$$

Roll is the rotation about the x axis that aligns the y and z components with gravity:

$$\phi_t = \text{atan2}(a_y, a_z) \quad (10)$$

Pitch is the rotation about the y axis, found by comparing the x component against the magnitude of the remaining two:

$$\theta_t = \text{atan2}\left(-a_x, \sqrt{a_y^2 + a_z^2}\right) \quad (11)$$

Since yaw cannot be recovered from gravity alone, it was fixed at $\psi_t = 0$. Unlike the gyroscope integrator, this estimate depends only on the current measurement, so each sample produces an orientation independently with no integration over time. These Euler angles were used to plot roll, pitch, and yaw against the Vicon measurements.

C. Complementary Filter

The gyroscope integrator tracks fast motion well but drifts as the bias accumulates, while the accelerometer estimate does not drift but is noisy in the short term and only works with low acceleration. The complementary filter combines the two so that each covers the other's weaknesses, using the high frequency information of the gyroscope and constraining the drift by using the absolute measurements of the accelerometer. At each step the gyroscope predicts the current orientation by propagating the previous prediction with the current measurement. This used the same equation from the gyroscope estimator as seen in equation (8). The accelerometer estimate Φ_t^a was determined using the methods previously described in equations (10) and (11).

The gyroscope estimates are converted to Euler angles (ϕ, θ, ψ) using xyz sequence and combined with the accelerometer Euler estimates in each axis by

$$\tilde{\Phi}_t = A\Phi_t^\omega + (I - A)\Phi_t^a \quad (12)$$

where $A = \text{diag}(\alpha, \alpha, 1)$, α is a tunable fusion factor set to 0.98, and I is the identity matrix. The filter follows the value of the gyroscope in the short term, rejecting the noise in the accelerometer and linear accelerations, while the $(1 - \alpha)$ term of the accelerometer consistently pulls the estimate toward the accelerometer estimate. Yaw is determined by the gyroscope only, since the accelerometer cannot observe the rotation about the gravity vector.

D. Madgwick Filter

The Madgwick filter [3] fuses the same two measurements as the complementary filter, but it performs the fusion directly on the quaternion rather than on Euler angles, and it poses the accelerometer correction as an optimization problem. Instead of blending two orientations, it blends two estimates of the *rate of change* of orientation: one obtained from the gyroscope, and another obtained from a gradient descent step that pulls the estimate toward the orientation implied by the direction of gravity. The fused rate is integrated onto the current quaternion, and the process repeats at every timestep.

The gyroscope measurement gives the quaternion derivative directly. Writing the angular velocity as a pure quaternion $(0, \vec{\omega}_t)$, the rate of change of orientation obtained from the

gyroscope is shown below. Integrating equation (13) is equivalent to the gyroscope integrator as described earlier, and drifts for the same reason.

$$\dot{q}_{\omega,t+1} = \frac{1}{2} \hat{q}_{est,t} \otimes (0, \vec{\omega}_{t+1}) \quad (13)$$

$$q_{\omega,t+1} = q_{est,t} + \dot{q}_{\omega,t+1} \Delta t \quad (14)$$

The accelerometer contribution is posed as an optimization problem. If the body is standing relatively still, the normalized measurement $\hat{\mathbf{a}}_t$ should coincide with the earth gravity direction $\hat{\mathbf{g}} = [0, 0, 0, 1]$ rotated into the sensor frame by the current orientation. The orientation that best aligns the gravity vector with the measurement is therefore the one that minimizes the difference between the two,

$$\min_{q \in \mathbb{R}^4} f(q, \hat{\mathbf{a}}_t) \quad (15)$$

$$f(q, \hat{\mathbf{a}}_t) = q^* \otimes \hat{\mathbf{g}} \otimes q - \hat{\mathbf{a}}_t \quad (16)$$

Equation (16) is the objective function that is used in the gradient descent algorithm. The gradient of this function gives the direction in which that difference grows fastest. Stepping against this direction moves the estimated orientation toward one that agrees with the measurement. The descent is defined by (17) for n iterations starting from an initial guess q_0 , with a step size μ [3], where the error direction on the solution surface is computed from the objective function and its Jacobian as in (18). We only require the error direction as the step size is determined by a fixed value μ rather than how far the estimate is from the actual value.

$$q_{n+1} = q_n - \mu \frac{\nabla f(q_n, \hat{\mathbf{a}}_t)}{\|\nabla f(q_n, \hat{\mathbf{a}}_t)\|}, \quad n = 0, 1, 2 \dots \quad (17)$$

$$\nabla f(q_n, \hat{\mathbf{a}}_t) = J^T(q_n) f(q_n, \hat{\mathbf{a}}_t) \quad (18)$$

Since gravity only acts along one axis of the earth frame, we can simplify the objective function f and its Jacobian J in (18) to (19) and (20) [3] respectively, where $q = [q_1 \ q_2 \ q_3 \ q_4]$ and q_1 is the scalar part of the quaternion.

$$f_g(q, \hat{\mathbf{a}}_t) = \begin{bmatrix} 2(q_2 q_4 - q_1 q_3) - a_x \\ 2(q_1 q_2 + q_3 q_4) - a_y \\ 2(\frac{1}{2} - q_2^2 - q_3^2) - a_z \end{bmatrix} \quad (19)$$

$$J_g(q) = \begin{bmatrix} -2q_3 & 2q_4 & -2q_1 & 2q_2 \\ 2q_2 & 2q_1 & 2q_4 & 2q_3 \\ 0 & -4q_2 & -4q_3 & 0 \end{bmatrix} \quad (20)$$

As mentioned in [3], it is acceptable to run one iteration of the gradient descent update step per time sample provided that the convergence rate governed by μ_t is greater than the rate of change of orientation. As long as this is true, we will eventually converge onto the solution given enough timesteps. Given the information above, we can rewrite (17) and (18) as

$$q_{\nabla,t+1} = \hat{q}_{est,t} - \mu_{t+1} \frac{\nabla f}{\|\nabla f\|} \quad (21)$$

$$\nabla f = J_g^T(\hat{q}_{est,t}) f_g(q_{est,t}, \hat{\mathbf{a}}_{t+1}) \quad (22)$$

Now that both estimates of orientation are available from (14) and (21), they can be fused with a weight γ_{t+1} applied to each, in the same manner as the fusion factor of the complementary filter,

$$\hat{q}_{est,t+1} = \gamma_{t+1} q_{\nabla,t+1} + (1 - \gamma_{t+1}) q_{\omega,t+1} \quad (23)$$

where $0 \leq \gamma_{t+1} \leq 1$. The best value of γ_{t+1} is the one for which the divergence of the gyroscope integration is exactly balanced by the convergence of the gradient descent algorithm [3]. Writing the divergence rate of $q_{\omega,t+1}$ as β and the convergence rate of $q_{\nabla,t+1}$ as $\mu_{t+1}/\Delta t$, this balance is expressed by (24), which rearranges to (25).

$$(1 - \gamma_{t+1})\beta = \gamma_{t+1} \frac{\mu_{t+1}}{\Delta t} \quad (24)$$

$$\gamma_{t+1} = \frac{\beta}{\frac{\mu_{t+1}}{\Delta t} + \beta} \quad (25)$$

The only requirement placed on μ_{t+1} is that it must be larger than the physical rate of change of orientation, and there is no upper bound on it. It can therefore be taken to be arbitrarily large, which simplifies both terms of (23). In (25) the β in the denominator becomes negligible, so γ_{t+1} reduces to (26) and can be approximated as zero. In the descent step, a very large μ_{t+1} makes the $\hat{q}_{est,t}$ term negligible against the step itself, so $q_{\nabla,t+1}$ reduces to (27).

$$\gamma_{t+1} \approx \frac{\beta \Delta t}{\mu_{t+1}} \approx 0 \quad (26)$$

$$q_{\nabla,t+1} \approx -\mu_{t+1} \frac{\nabla f}{\|\nabla f\|} \quad (27)$$

Substituting (26) and (27) into (23), and writing the gyroscope estimate as (14), the fusion step simplifies into the two equations written below. Note that γ_{t+1} is substituted as $\beta \Delta t / \mu_{t+1}$ in the first term of (23), where it cancels the μ_{t+1} introduced by (27), and as zero in the second.

$$\dot{q}_{est,t+1} = \dot{q}_{\omega,t+1} - \beta \frac{\nabla f}{\|\nabla f\|} \quad (28)$$

$$q_{est,t+1} = \hat{q}_{est,t} + \dot{q}_{est,t+1} \Delta t \quad (29)$$

Equation (28) and (29) are the two main results of the Madgwick filter that allow us to estimate the current state quaternion from the previous state quaternion and gyroscope and accelerometer measurements. The only unknown quantity is β , which is a tunable parameter. We found $\beta = 0.2$ to work best on the datasets provided to us.

Equations (28) and (29) were executed at every timestep to compute the orientation estimate. Unlike the gyroscope integrator, the estimate was initialized to the identity quaternion $[1, 0, 0, 0]$, since the gradient descent term corrects the initial condition in roll and pitch, but not yaw, on its own. Finally, the resulting quaternions were converted to Euler angles to plot roll, pitch, and yaw against the Vicon measurements.

E. Unscented Kalman Filter

The complementary and Madgwick filters both return a single orientation at every timestep, with no indication of how much that orientation should be trusted. The balance they strike between the gyroscope and the accelerometer is governed by a manually tuned constant, so it cannot react when the conditions change. A Bayesian filter instead carries a probability distribution over the orientation, which tracks the estimate and its uncertainty together. The Kalman filter is the simplest of these. It assumes linear process and measurement models with Gaussian noise, and since a linear function of a Gaussian is still Gaussian, the entire distribution can be represented by just a mean and a covariance. Almost nothing in the real world is linear, so this is restrictive. The extended Kalman filter gets around it by linearizing both models with a first order Taylor expansion about the current estimate. This requires Jacobians with respect to the state and the noise, which are tedious to derive and are not guaranteed to exist. The unscented Kalman filter (UKF) keeps the Gaussian assumption but drops the linear one. Instead of linearizing, it uses a small and deliberately chosen set of points, called sigma points, whose mean and covariance match those of the current state distribution. Each point is passed through the nonlinear model, and the transformed points are then used to calculate the new mean and covariance. This generally gives a better estimate than approximating the model as linear. The idea behind it is that approximating a distribution is easier than approximating an arbitrary nonlinear function.

1) *State Vector and Covariance*: The state estimate x is a 7-dimensional vector, where $x = [q, \vec{\omega}]$. Here q is the orientation of the body expressed in the world frame, and $\vec{\omega}$ is the angular velocity of the body expressed in the body frame. The state has 7 elements but only 6 degrees of freedom, since q must be a unit quaternion, so the covariance P is a 6×6 matrix [4]. Its six entries track the error in the orientation, written as a rotation vector, and the error in the angular velocity. We initialize P to be

$$P_0 = \begin{bmatrix} \sigma_q^2 I_3 & 0_{3 \times 3} \\ 0_{3 \times 3} & \sigma_\omega^2 I_3 \end{bmatrix} \quad (30)$$

where I_3 is the 3×3 identity matrix, σ_q is the initial standard deviation of the orientation, assumed to be 10° , and σ_ω is the initial standard deviation of the angular velocity, assumed to be $20^\circ/\text{s}$.

2) *Generating the Sigma Points*: The first step of every UKF recursion is to transform the previous state estimate x_t and its covariance P_t into a set of sigma points [4]. We obtain these points from the Cholesky decomposition of the previous covariance,

$$P_t = SS^\top \quad (31)$$

where each column of S gives a direction and a distance away from the mean. These columns span the same covariance ellipsoid as P_t , but they need not be its orthogonal principal axes. Any set of vectors that reproduces the ellipsoid will do, and the Cholesky decomposition is cheaper to compute than a full eigendecomposition.

Each of the n columns of S is then scaled by $\pm\sqrt{n}$ to form the set $\{\mathcal{W}_i\}$,

$$\mathcal{W}_{i,i+n} = \pm\sqrt{n} \text{col}_i(S), \quad i = 1, \dots, n \quad (32)$$

where $n = 6$, so the set holds $2n = 12$ points. $\{\mathcal{W}_i\}$ has zero mean, since the positive and negative pairs cancel each other out, and the same covariance as P_t .

The mean is then shifted onto the previous estimate. For an ordinary state vector this would simply be $x_t + \mathcal{W}_i$, but that is not possible here, because the orientation part of x_t is a quaternion while the corresponding part of $\mathcal{W}_i$ is a rotation vector. Writing $\mathcal{W}_i = [\vec{w}_{q,i}, \vec{w}_{\omega,i}]$, we convert $\vec{w}_{q,i}$ into a quaternion $q_{w,i}$ using the same axis-angle conversion as in (8), taking the angle as $\|\vec{w}_{q,i}\|$ and the axis as its direction, and apply it to the current orientation by quaternion multiplication. The angular velocity is unconstrained, so it is added directly. This is the step that moves from the six-dimensional $\{\mathcal{W}_i\}$ to the seven-dimensional sigma points [4].

$$\mathcal{X}_i = \begin{bmatrix} q_{w,i} \otimes q_t \\ \vec{\omega}_t + \vec{w}_{\omega,i} \end{bmatrix} \quad (33)$$

This gives a set $\{\mathcal{X}_i\}$ of $2n$ sigma points with mean x_t and covariance P_t . The process noise is applied to these points before they are passed through the process model rather than after. Disturbing a state vector by the process noise takes the same form as (33), so the two steps can be combined by adding the process noise covariance Q to P_t before the sigma points are generated [4],

$$P_t + Q = SS^\top \quad (34)$$

and the S of (34) is used in (32) in place of the one from (31). The process noise is therefore accounted for entirely by the spread of the sigma points, and does not appear anywhere else in the filter. We set $Q = 10^{-4}I_6$, which works with the measurement-noise covariance R to control how the filter balances the process prediction and IMU measurements.

3) *Process Model*: After the sigma points $\{\mathcal{X}_i\}$ are obtained, the process model A is used to project each point ahead in time, resulting in a different set of state vectors $\{\mathcal{Y}_i\}$ [4]. Writing the components of a sigma point as $\mathcal{X}_i = [q_i, \vec{\omega}_i]$, every point is propagated on its own, using its own angular velocity,

$$\mathcal{Y}_i = A(\mathcal{X}_i, 0) = \begin{bmatrix} q_i \otimes q_{\Delta,i} \\ \vec{\omega}_i \end{bmatrix} \quad (35)$$

where $q_{\Delta,i}$ is the quaternion built from the angle and axis of (5) and (6) evaluated at $\vec{\omega}_i$. This is the same propagation as the gyroscope integrator in (8), except that the angular velocity is taken from the sigma point rather than from the measurement. The model assumes the angular velocity is constant over the timestep, so it is carried through unchanged.

The second argument of A is zero because the process noise has already been accounted for in the spread of the sigma points in (34), and so is not applied a second time here [4].

4) *Transformed Mean and Covariance:* If the set of transformed state vectors $\{\mathcal{Y}_i\}$ were elements of a vector space, we could simply calculate the barycentric mean of the set [4],

$$\bar{x} = \frac{1}{2n} \sum_{i=1}^{2n} \mathcal{Y}_i \quad (36)$$

This is valid for the angular velocity component, so the mean angular velocity $\vec{\omega}$ was taken directly from (36). It is not valid for the quaternion component, because orientations are periodic, and the mean of two unit quaternions is not necessarily a unit quaternion [4]. To solve this problem we use the intrinsic gradient descent algorithm described in [5].

The mean quaternion is estimated one iteration at a time, with the iteration written as j . Writing the quaternion component of $\mathcal{Y}_i$ in (35) as q_i^- , in each iteration we compute error vectors between the current estimate $\bar{q}_j$ and every q_i^- in the set. The quaternion representation of the error vector is

$$e_i = q_i^- \otimes \bar{q}_j^{-1} \quad (37)$$

which is the difference in rotation between the estimated mean and the set element. Each e_i is converted to its axis-angle vector representation $\vec{e}_i$, and the barycentric mean of these error vectors is calculated,

$$\bar{e} = \frac{1}{2n} \sum_{i=1}^{2n} \vec{e}_i \quad (38)$$

This mean is a measure of how far the current estimate is from the true mean, and it also points in the direction of the true mean. The same rotation vector can therefore be used to update the estimate for the next iteration,

$$\bar{q}_{j+1} = \bar{e} \otimes \bar{q}_j \quad (39)$$

where $\bar{e}$ is the quaternion form of $\bar{e}$. We continue until the magnitude of the mean error vector reaches the desired accuracy, which we set at $\|\bar{e}\| < 10^{-3}$, with the number of iterations capped at 100. The starting value is arbitrary but determines how many iterations are needed, so to reduce that number we initialized $\bar{q}_0$ with the previous state estimate quaternion q_i . Now that we have the mean of $\{\mathcal{Y}_i\}$, we can find its covariance using

$$P_{t+1}^- = \frac{1}{2n} \sum_{i=1}^{2n} \mathcal{W}_i' (\mathcal{W}_i')^\top \quad (40)$$

where $\mathcal{W}_i'$ is the difference between the transformed sigma point and the transformed mean, written as a six-dimensional vector in the same way as $\mathcal{W}_i$ in (32). For the angular velocity part this is an ordinary subtraction, $\vec{\omega}_i - \vec{\omega}$. For the quaternion part it is the difference in rotation between the mean and the set element, which we already have as the $\vec{e}_i$ from the last iteration of the gradient descent algorithm [4], so no extra work is needed. The covariance of $\{\mathcal{Y}_i\}$ is therefore built from

$$\mathcal{W}_i' = \begin{bmatrix} \vec{e}_i \\ \vec{\omega}_i - \vec{\omega} \end{bmatrix} \quad (41)$$

The converged $\bar{q}$ together with $\vec{\omega}$ forms the *a priori* state estimate $x_{t+1}^- = [\bar{q}, \vec{\omega}]$, and P_{t+1}^- from (40) is its covariance. Both are corrected by the measurement update step.

5) *Measurement Model:* Before we get to the measurement update step of the filter, we must first define the measurement model H , which maps each transformed sigma point to the measurement we would expect to observe from it,

$$\mathcal{Z}_i = H(\mathcal{Y}_i, 0) = \begin{bmatrix} \vec{\omega}_i \\ (q_i^-)^{-1} \otimes \mathbf{g} \otimes q_i^- \end{bmatrix} \quad (42)$$

where $\mathbf{g} = (0, 0, 0, 9.81)$ is the gravity vector in the world frame, written as a pure quaternion. Each $\mathcal{Z}_i$ holds the gyroscope reading followed by the accelerometer reading, in the same order as the actual measurement z_{t+1} . The gyroscope is observed directly, since the angular velocity is already part of the state, so the first three components are simply the angular velocity of the sigma point. For the accelerometer we use the quaternion part of $\mathcal{Y}_i$ to rotate the gravity vector from the world frame into the body frame. This carries the same assumption as the accelerometer estimator, that the IMU undergoes little linear acceleration, so that the accelerometer measures only the reaction to gravity.

The second argument of H is zero because the measurement noise is not applied to the sigma points. Unlike the process noise, which was folded into their spread in (34), the measurement noise covariance R enters later as part of the innovation covariance in the measurement update step.

6) *Measurement Update:* Both the gyroscope and accelerometer components of $\{\mathcal{Z}_i\}$ are elements of a vector space, so we can simply find the barycentric mean and the covariance of the set,

$$\bar{z}_{t+1} = \frac{1}{2n} \sum_{i=1}^{2n} \mathcal{Z}_i \quad (43)$$

$$P_{zz} = \frac{1}{2n} \sum_{i=1}^{2n} (\mathcal{Z}_i - \bar{z}_{t+1}) (\mathcal{Z}_i - \bar{z}_{t+1})^\top \quad (44)$$

The innovation v_{t+1} is the difference between the actual measurement z_{t+1} and its predicted value $\bar{z}_{t+1}$ [4]. Its covariance is the sum of P_{zz} , the uncertainty in the measurement caused by the uncertainty in the state vector prediction, and the measurement noise covariance R [4].

$$v_{t+1} = z_{t+1} - \bar{z}_{t+1} \quad (45)$$

$$P_{vv} = P_{zz} + R \quad (46)$$

Here R has the same block diagonal form as P_0 in (30), with a standard deviation of $3^\circ/\text{s}$ on the gyroscope measurements and 0.5 m/s^2 on the accelerometer measurements.

The cross correlation between the state and the measurement is built using the same six-dimensional errors $\mathcal{W}_i'$ defined in (41) that was used for the *a priori* covariance, so no new quantity is needed,

$$P_{xz} = \frac{1}{2n} \sum_{i=1}^{2n} \mathcal{W}_i' (\mathcal{Z}_i - \bar{z}_{t+1})^\top \quad (47)$$

and the Kalman gain is given by

$$K_{t+1} = P_{xz} P_{vv}^{-1} \quad (48)$$

The correction applied to the *a priori* estimate is the innovation scaled by the gain,

$$\delta_{t+1} = K_{t+1} v_{t+1} \quad (49)$$

Like $\mathcal{W}_i$, δ_{t+1} is six-dimensional, so it splits into a rotation vector part $\vec{\delta}_q$ and an angular velocity part $\vec{\delta}_\omega$. The rotation vector is converted to a quaternion q_δ and applied to the *a priori* orientation by quaternion multiplication, while the angular velocity is added directly.

$$x_{t+1} = \begin{bmatrix} q_\delta \otimes \bar{q} \\ \vec{\omega} + \vec{\delta}_\omega \end{bmatrix} \quad (50)$$

Finally, the covariance is corrected as

$$P_{t+1} = P_{t+1}^- - K_{t+1} P_{vv} K_{t+1}^\top \quad (51)$$

Apart from the initialization of the state and its covariance, every step above was repeated at each timestep, with x_{t+1} and P_{t+1} becoming the inputs to the next recursion. As with the other filters, the resulting quaternions were converted to Euler angles to plot roll, pitch, and yaw against the Vicon measurements.

IV. RESULTS

Each filter was run on all 6 training sequences and compared against the time-aligned Vicon orientation. The timestamps between the data and the Vicon system were aligned using a custom SLERP algorithm written by Claude Opus 5 [1]. The orientation error was determined using the minimum distance between the reference quaternions and the estimated quaternions, which avoids the wraparound artifacts of Euler angles. The per-axis error is the xyz Euler decomposition of the relative rotation $R_k^{\text{err}} = (R_k^{\text{vic}})^\top R_k^{\text{est}}$, giving roll, pitch, and yaw components. For every quantity the root-mean-square value over the trajectory was reported,

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N e_k^2}. \quad (52)$$

Table I lists the total RMSE for each filter on each data set, and Table II gives the per-axis RMSE averaged over the six training sequences. Plots of the estimated and reference orientations, whenever available, for all datasets are provided at the end of the report.

The gyroscope integrator has no absolute reference to prevent drift, so any remaining gyroscope bias accumulates over time. Among the three axes, it achieved lowest average RMSE in pitch and its highest in yaw. The accelerometer estimate does not accumulate integration drift, but it becomes noisy during linear motion and cannot observe yaw. When a dataset has a large movement in yaw, it performs badly in the tests due to that fact, such as datasets 3, 5, and 6, while the other sequences have a relatively low residual.

The complementary filter corrects the roll and pitch drift from the gyroscope integrator using the accelerometer measurements, while the yaw estimate still comes entirely from the gyroscope. This improves the pitch estimate to an average RMSE of 8.3° , compared to 10.3° for the gyroscope integrator. The filter performs well on datasets 3 and 6 with total RMSE values of 4.0° and 6.8° , but performs poorly on dataset 1 with an RMSE of 77.6° .

The Madgwick filter achieved a mean total RMSE of 18.5° , which was lower than the gyroscope integrator, accelerometer estimator, and complementary filter, but higher than the UKF. Its average roll, pitch, and yaw errors were 6.6° , 5.9° , and 16.2° , respectively. The filter can recover from an incorrect initial orientation in roll and pitch because the accelerometer provides a reference to gravity. However, the accelerometer provides no reference for yaw, so an incorrect initial yaw remains as a constant offset throughout the estimate.

The unscented Kalman filter performed the best overall, achieving the lowest mean total RMSE of 14.6° . It performed particularly well on datasets 3 and 6, with total RMSE values of 4.9° and 5.1° , respectively. Dataset 4 was its weakest, with a total RMSE value of 34.1° , although this was still lower than the errors produced by most other filters on this sequence. The UKF also achieved the lowest average error in all 3 axes, with RMSE values of 5.4° for roll, 4.8° for pitch, and 12.5° for yaw. Its yaw error remained higher because the accelerometer provides no absolute reference for heading.

A. Rotation Animations

For each dataset provided, the *rotplot* function was used to visually show the performance of the gyroscope integrator, accelerometer estimator, complementary filter, Madgwick filter, and unscented Kalman filter alongside the Vicon ground truth data, whenever available. The gyroscope integrator drifts but follows the reference closely at the start. The accelerometer estimator is very jittery, showing the noise in the estimates and its inability to follow the heading. The complementary filter and the Madgwick filter track the reference very well. The unscented Kalman filter tracks the reference the most closely of the five, and drifts the least in heading. The videos can be found here: <https://youtu.be/cGSoPIDG6GQ>.

V. CONCLUSION

This project compared five attitude estimation methods using measurements from a 6-DoF IMU: a gyroscope integrator, an accelerometer tilt estimator, a complementary filter, a Madgwick filter, and an unscented Kalman filter. The methods were evaluated against Vicon motion-capture data across six training sequences and applied to four additional test sequences without ground truth. Each of the single-sensor estimators showed important limitations. The gyroscope integrator performs well for fast movements in the short term; however, it drifts over time and is not suitable for longer movements. It also requires an initial reference, since it can only track the change in orientation, not the orientation itself. The accelerometer estimator is able to track a global reference

TABLE I
TOTAL ORIENTATION RMSE IN DEGREES FOR EACH SEQUENCE

Filter	1	2	3	4	5	6	Mean
Gyroscope integrator	19.6	25.8	12.6	43.6	30.1	13.4	24.1
Accelerometer estimator	9.0	9.3	43.8	16.4	116.7	60.5	42.6
Complementary filter	77.6	7.3	4.0	43.2	20.0	6.8	26.5
Madgwick filter	13.3	14.5	15.0	44.6	17.3	6.3	18.5
Unscented Kalman filter	13.5	14.2	4.9	34.1	15.8	5.1	14.6

and thus does not drift. However, its measurements can be noisy, and further processing amplifies that noise, making it unsuitable for estimation on a drone. It also cannot determine yaw, since its measurements are derived from the gravity vector, which provides no information about rotation around that axis. This is reflected in its yaw RMSE of 38.4°, far higher than its roll (15.5°) and pitch (10.9°) errors.

The complementary filter combines the two approaches to leverage the strengths of each, achieving a mean total RMSE of 26.5° across the six sequences. Its best performance was on pitch (8.3°), followed by roll (13.8°), with yaw remaining the weakest axis (24.0°), since the accelerometer cannot assist with yaw correction. The Madgwick filter achieved the best performance of the four non-Bayesian methods, with a mean total RMSE of 18.5°. It performed particularly well on roll and pitch (6.6° and 5.9°, respectively), as it constrains its predictions to a feasible rotation by remaining on the quaternion manifold. Its yaw error (16.2°) was also the lowest of those four, though still noticeably higher than its roll and pitch performance.

The unscented Kalman filter achieved the best overall performance, with a mean total RMSE of 14.6°. It also produced the lowest average error in all three axes, with RMSE values of 5.4° in roll, 4.8° in pitch, and 12.5° in yaw. By using sigma points to represent a set of possible orientations and angular velocities at each timestep, the UKF was able to combine the gyroscope and accelerometer measurements while keeping track of uncertainties in the estimate. However, its yaw error remained higher than its roll and pitch errors because the accelerometer provides no absolute reference. Incorporating a magnetometer could provide this reference and further reduce long-term yaw drift.

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TABLE II
PER-AXIS RMSE IN DEGREES, AVERAGED OVER THE SIX SEQUENCES

Filter	Roll	Pitch	Yaw
Gyroscope integrator	13.1	10.3	16.8
Accelerometer estimator	15.5	10.9	38.4
Complementary filter	13.8	8.3	24.0
Madgwick filter	6.6	5.9	16.2
Unscented Kalman filter	5.4	4.8	12.5

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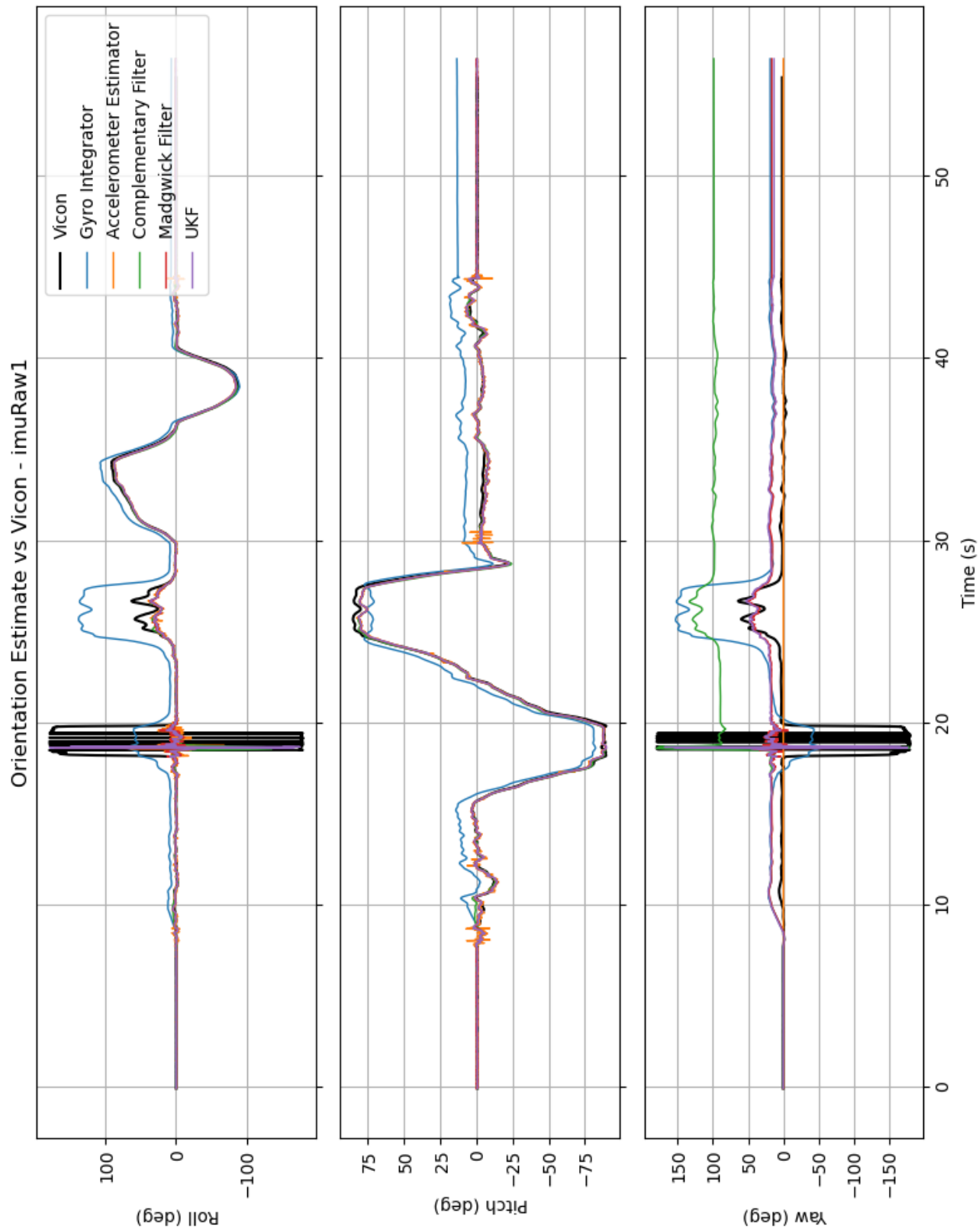


Fig. 1. Orientation estimates compared with Vicon measurements for Train Dataset 1.

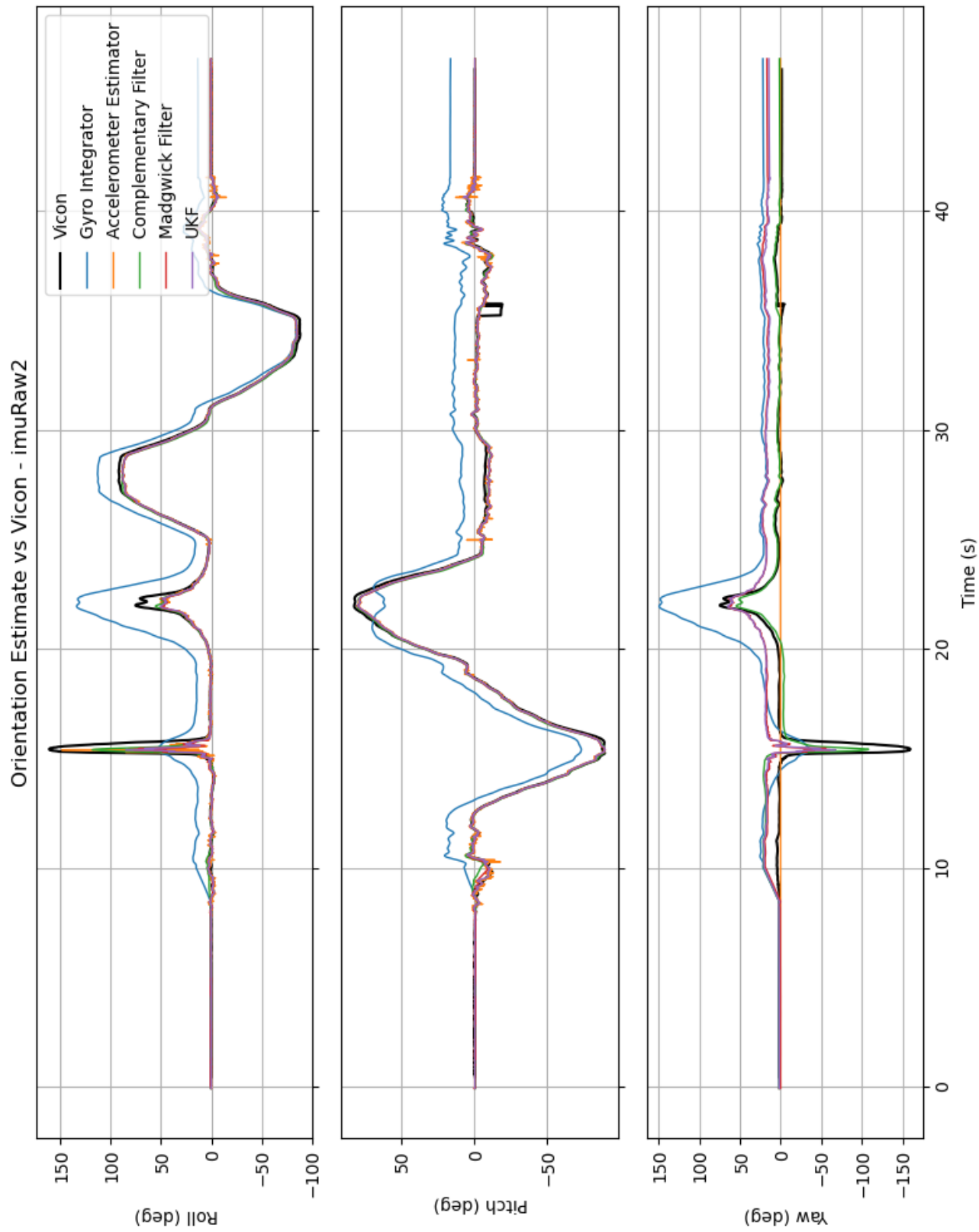


Fig. 2. Orientation estimates compared with Vicon measurements for Train Dataset 2.

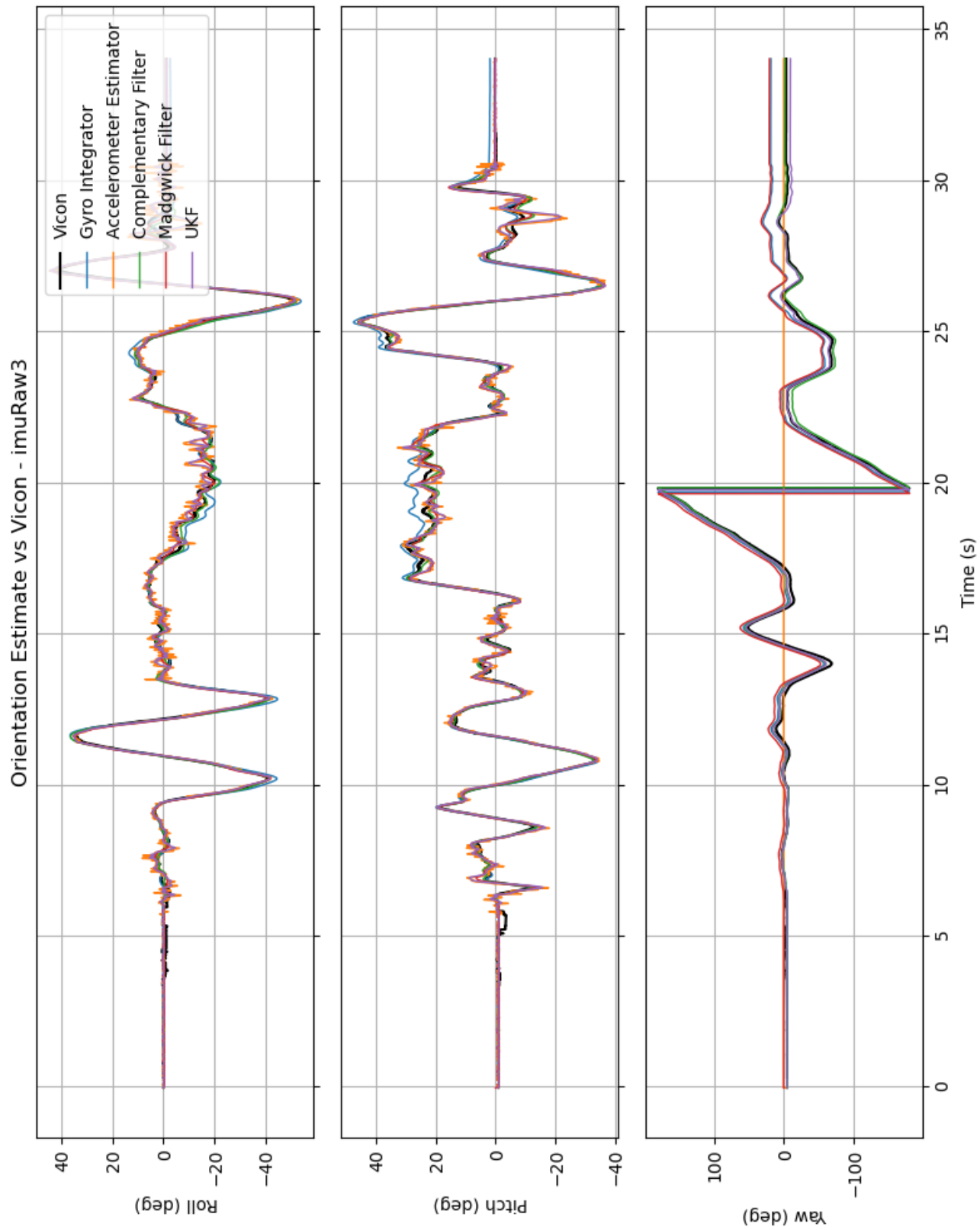


Fig. 3. Orientation estimates compared with Vicon measurements for Train Dataset 3.

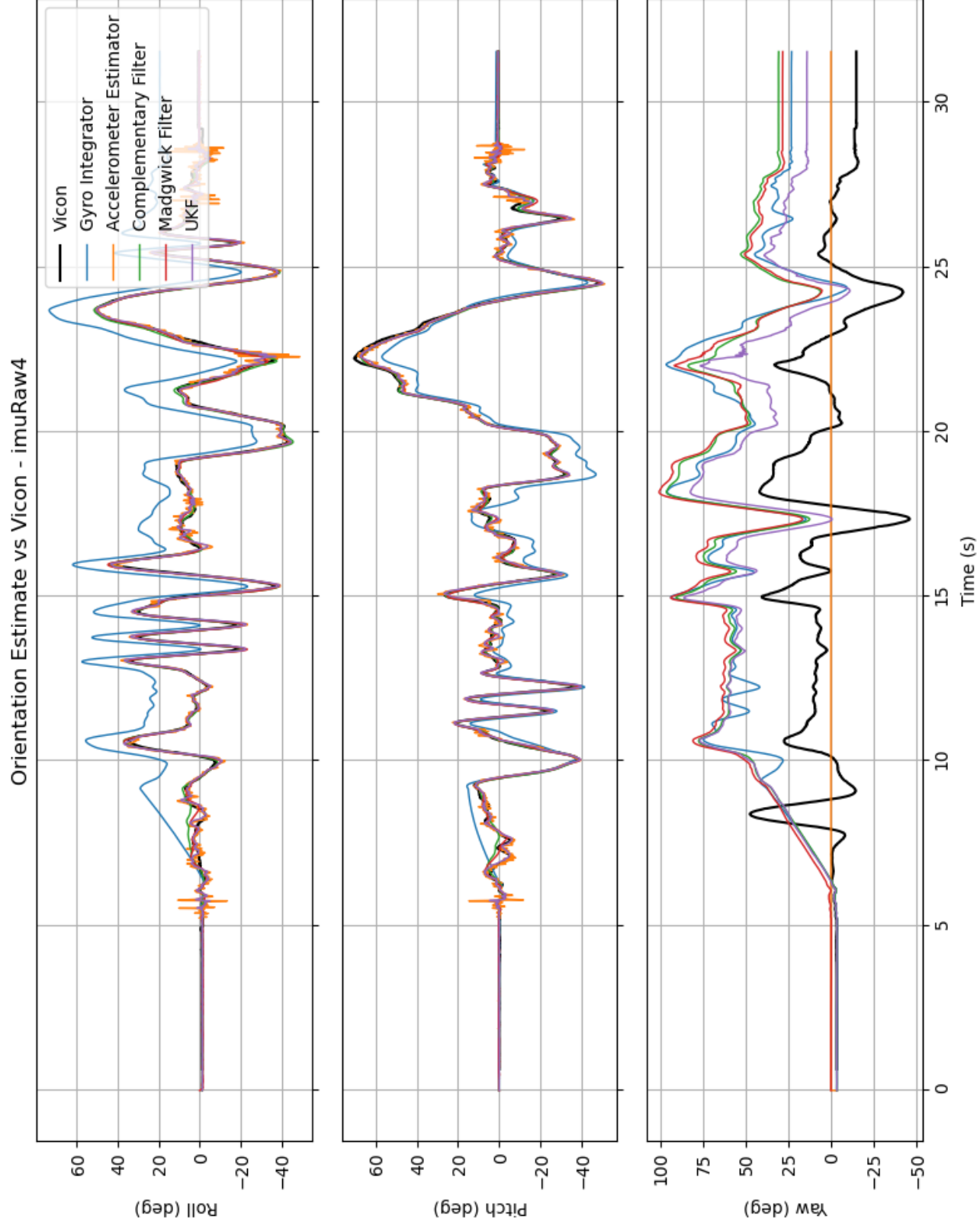


Fig. 4. Orientation estimates compared with Vicon measurements for Train Dataset 4.

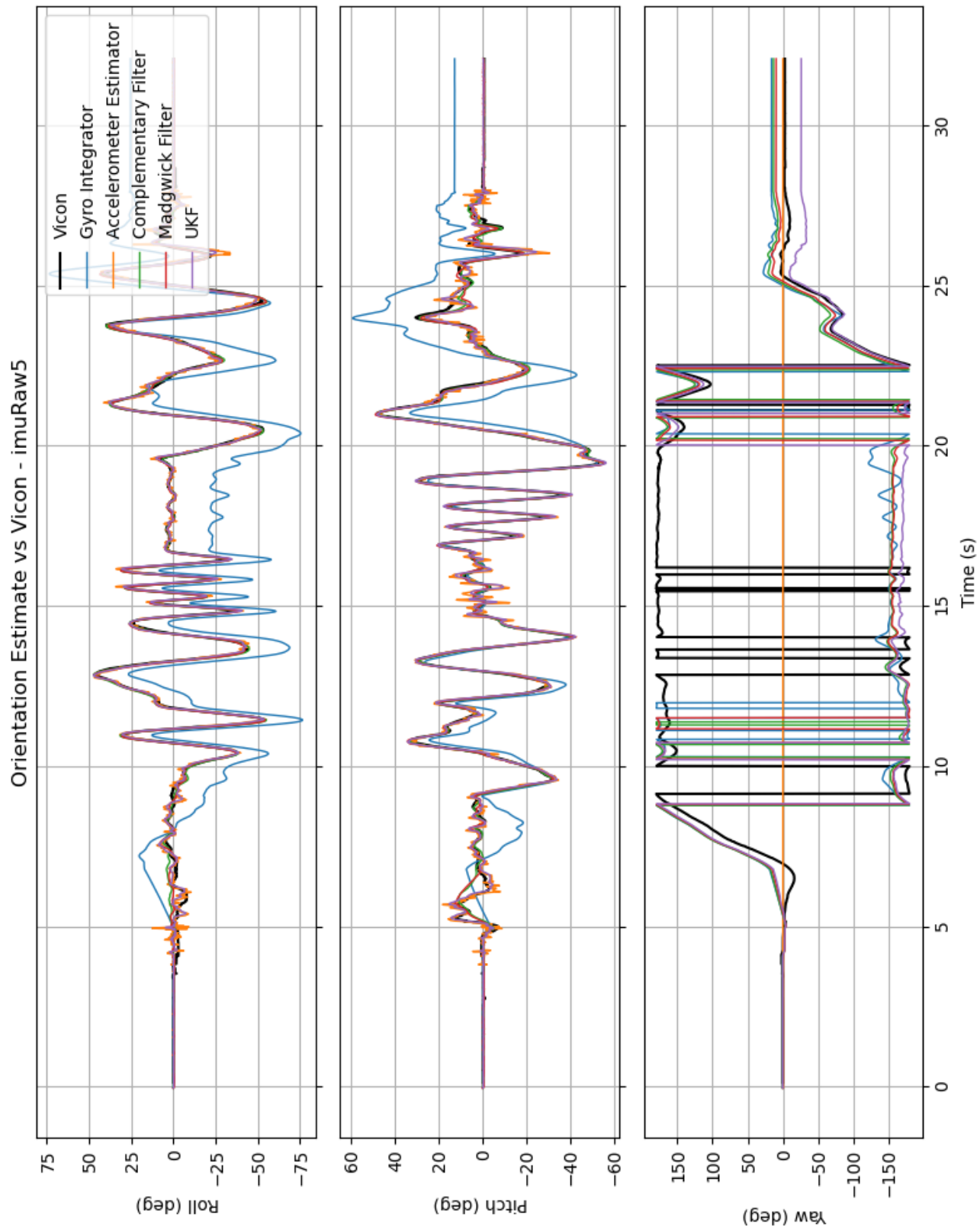


Fig. 5. Orientation estimates compared with Vicon measurements for Train Dataset 5.

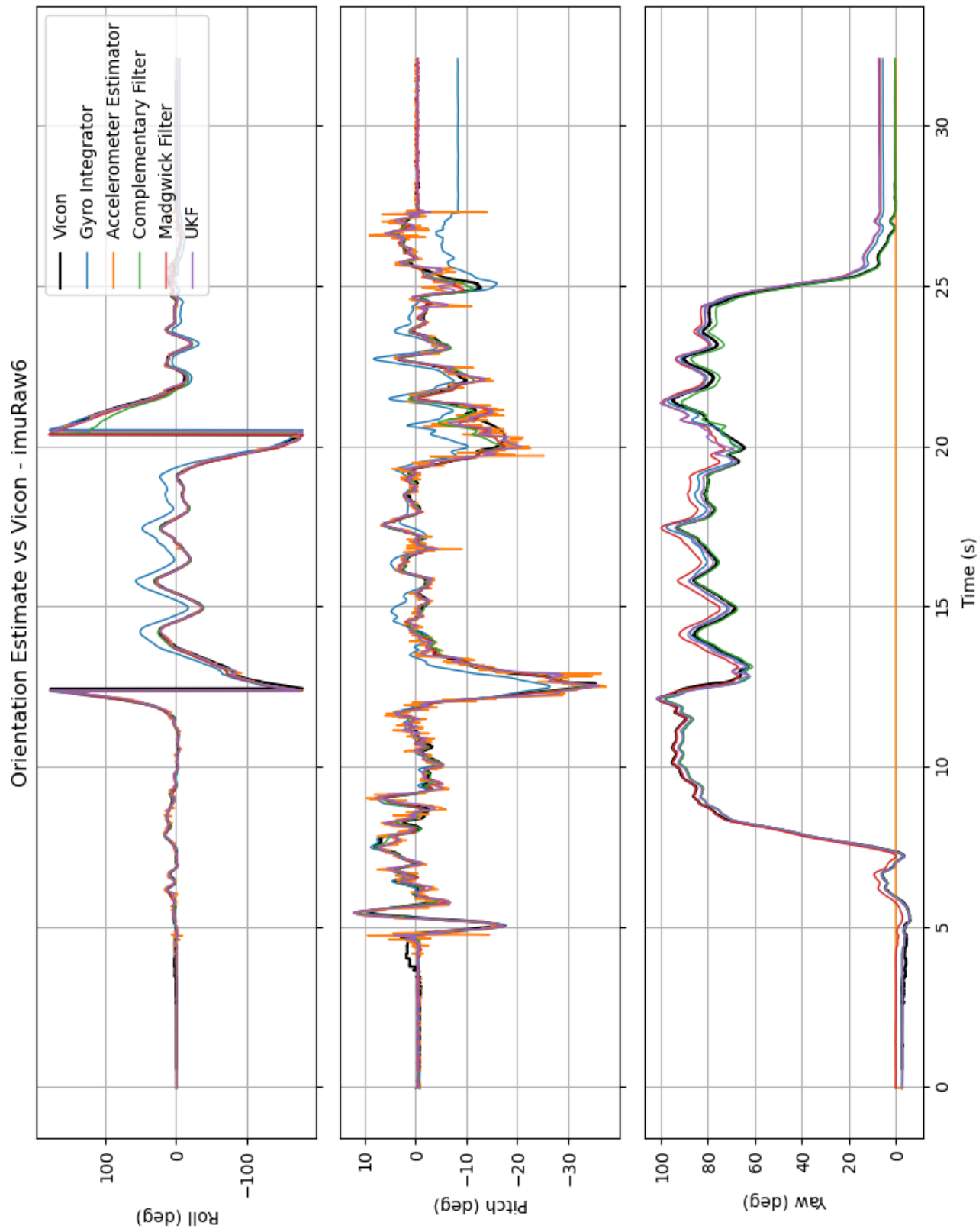


Fig. 6. Orientation estimates compared with Vicon measurements for Train Dataset 6.

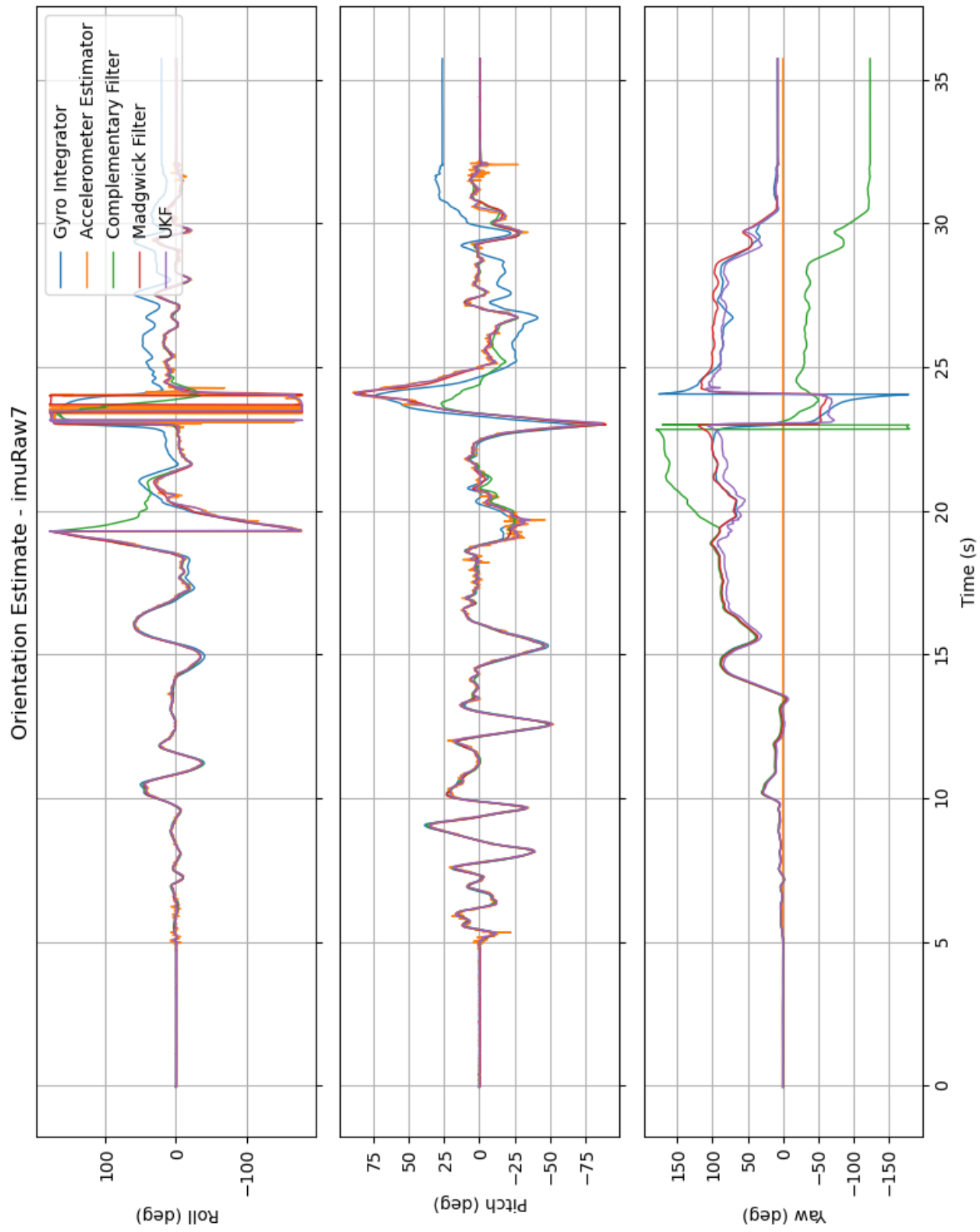


Fig. 7. Orientation estimates for the Test Dataset 7.

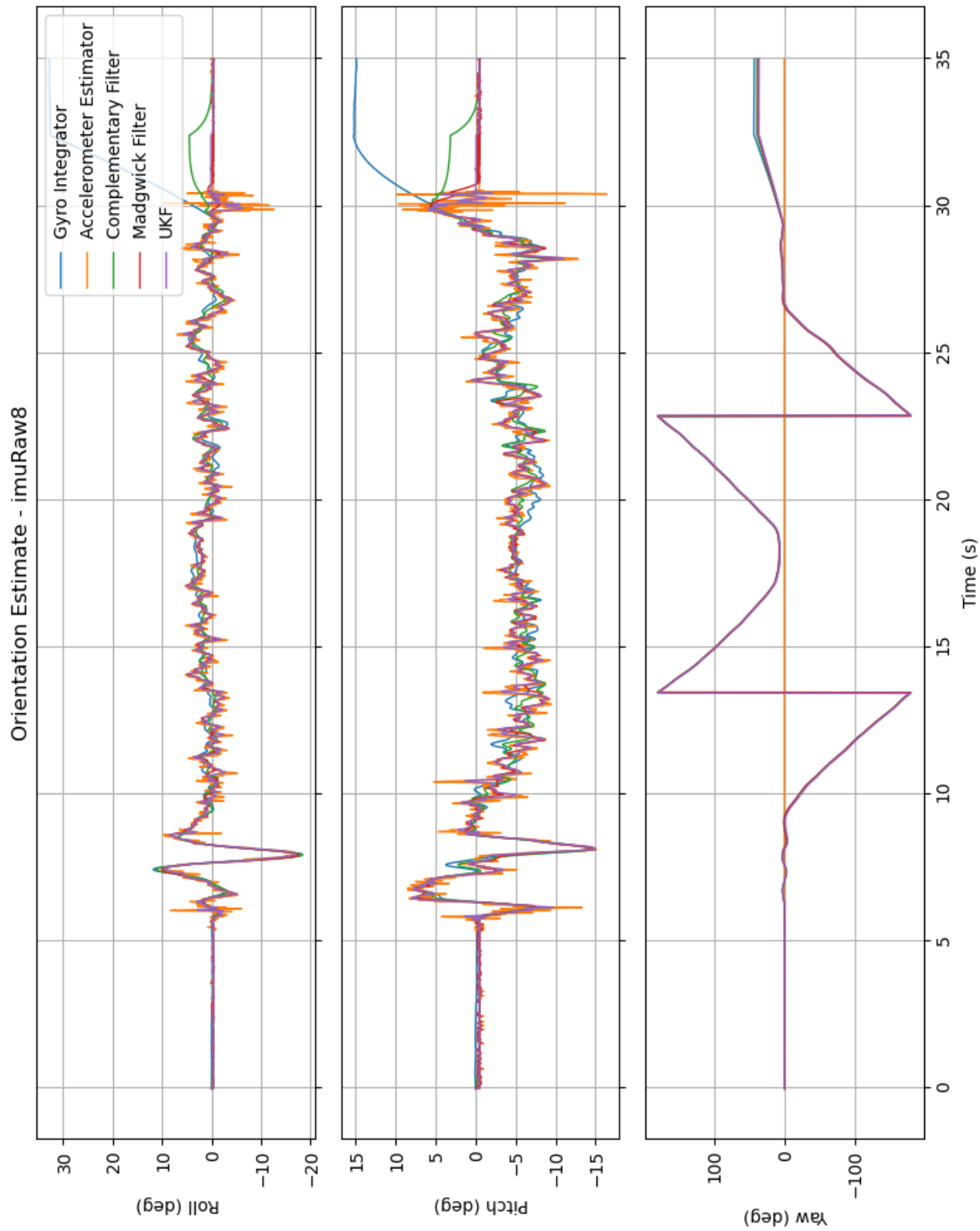


Fig. 8. Orientation estimates for the Test Dataset 8.

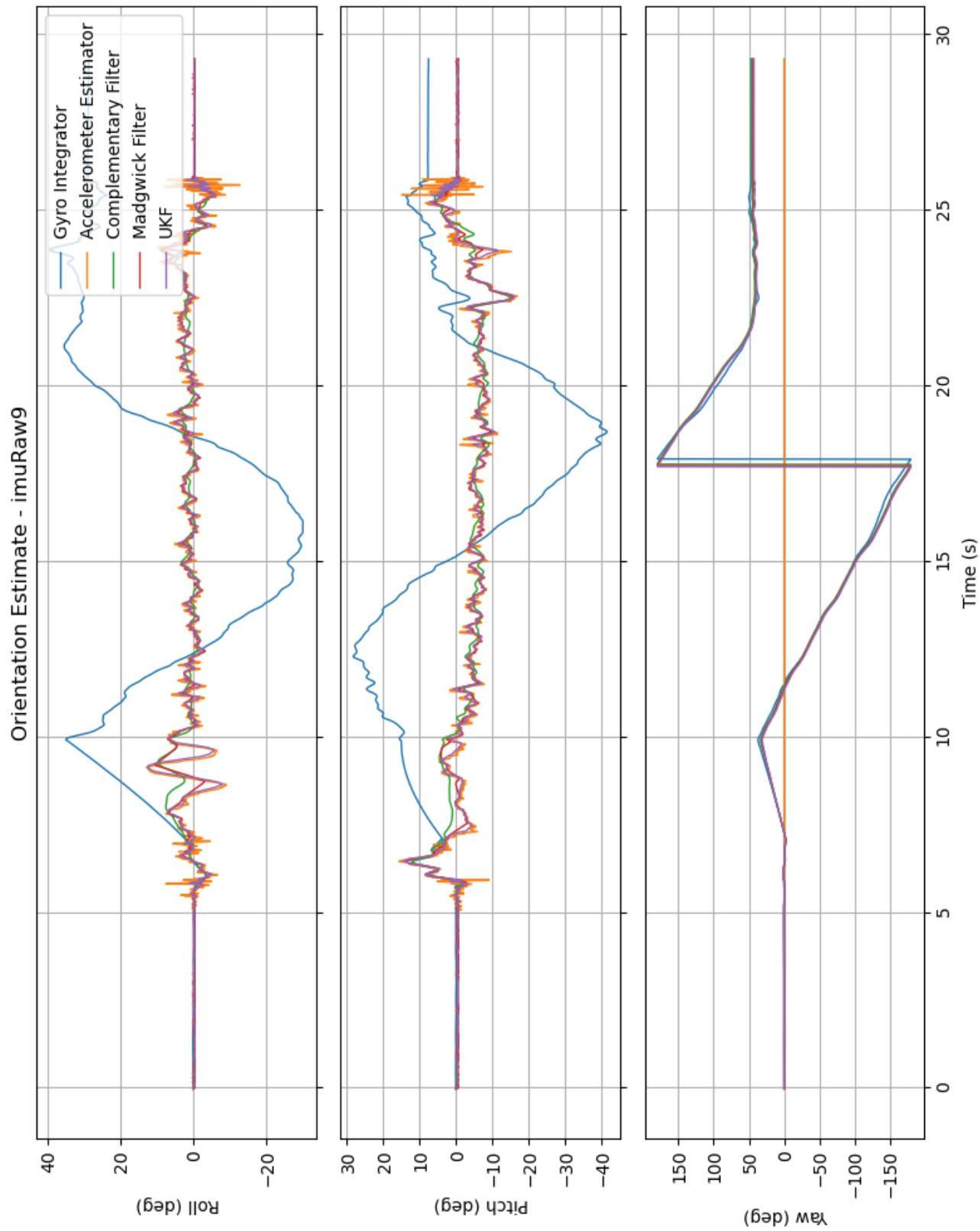


Fig. 9. Orientation estimates for the Test Dataset 9.

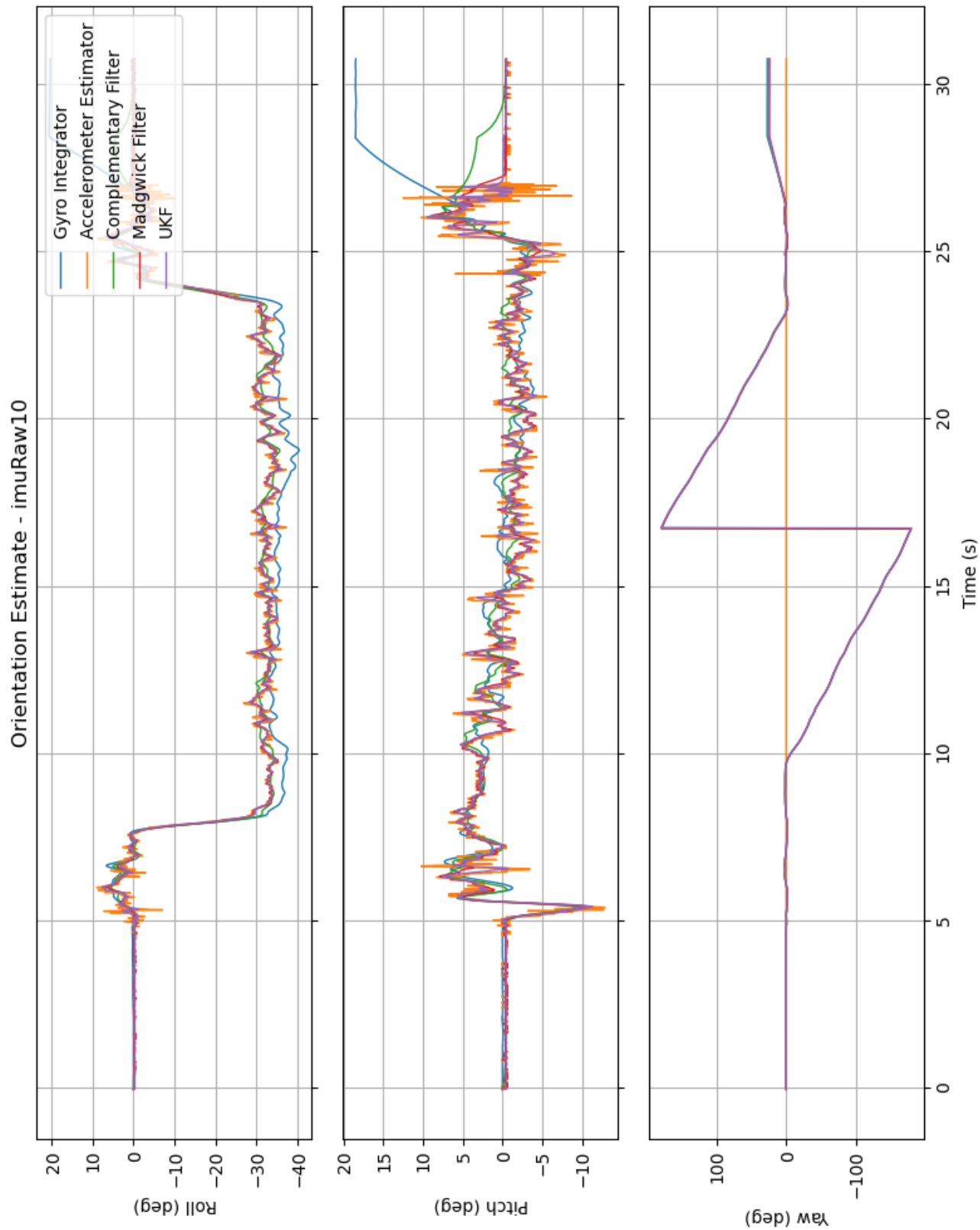


Fig. 10. Orientation estimates for the Test Dataset 10.